MIL-E-1/1228C(NAVY)
13 January 1967
SUPERSEDING
MIL-E-1/1228B(NAVY)
29 November 1962
(See Note 23)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TYPE 8361

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Magnetically focused pulse amplifier klystron

Dimensions	- See figures 1, 2 and 3
Base	- Special rigid leads
Cathode	- Coated unipotential
Weight	- 140 lbs. max.
Mounting position	- Any (Notes 12(a) & 14)
Ground	- Tube shell
Thermal interlocks	- Note 15
Cooling	- Liquid (Note 5)
Tuning	- Note 4

ABSOLUTE RATINGS:

Parameter:	Ef	peak beam voltage	peak inverse beam voltage	average beam current	peak beam current
Unit:	V	kv	kv	mA	a
Maximum:	5.25	135	50	225	111
Minimum:	4.25	---	--	---	---
	Note 1	Notes 2, 12(c) & 14		Note 22	Note 7

ABSOLUTE RATINGS:

Parameter:	pi	pi	Pd	pd	Alt	Po	po	Beam Du
Unit:	kw	Mw	W	kw	ft	kW	Mw	---
Maximum:	24.1	15	50	15	10,000	5.25	3.3	0.002
Minimum:	---	--	--	Note 11	---	---	---	---
	Note 22				Note 5(d)		Notes 8 & 10(a), (b), (c)	Note 6(d)

ABSOLUTE RATINGS:

Parameter:	Output r.f. Du	Beam tp	Output r.f. tp	Output Load VSWR	Output waveguide pressure
Unit:	---	us	us	---	psia
Maximum:	0.0016	5	3.2	1.7:1	65
Minimum:	---	--	---	---	45
	Note 6(d)	Note 6(a, b)	Note 6(c)	Note 10(d)	Note 10(d)

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ABSOLUTE RATINGS:

Parameter:	TA	Magnet coil current	Coolant pressure	Applied tuner torque	Cooling
Unit:	°C	amperes	psia	in-lbs.	°C
Maximum:	85	15	165	10	77
Minimum:	-32	--	---	--	10
	Note 5(e)	Note 12	Note 5(d)	Note 4(c)	Note 5

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - required (Note 21)

5. Preparation for delivery

E-50.2 Holding period - t = 168 hrs. (min)

METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Qualification					
4104	Mechanical fatigue	All tuners; No voltages; Note 19	---	1,000	---	cycles
1023	Temperature coefficient	po 2; Note 18	Δpo/ΔT	---	0.025	db/°C
4226	Warm-up time: po 2	Notes 11(b) and 13				
----	Low level gain	Ef = 4.75V; peak beam voltage = 128 kv; Magnet coil current = 11A; F = 5400 to 5900 Mc; Notes 6(b), (c), (d), 8(a), 9, 10(d), and 12(b), (d)	---	36	42	db
----	Shock	Nonoperating; Note 20				
1031	Vibration	Note 16				
----	Power output (4)	peak beam voltage = 128 kv; Ef = 4.75V; F = 5400 Mc, 5650 Mc, and 5900 Mc; pd = 10 kw; Magnet coil current = 12A; Notes 6(b)(c)(d), 8(a)(c)(e), 10(d) and 12(b)(d)				
		VSWR = 1.2:1 all phases	po 4a	3	---	Mw
		VSWR = 1.5:1 all phases;	po 4B	2.6	---	Mw
		VSWR = 1.7:1 all phases; output waveguide pressure of 45 to 65 psia		No damages		
1136	Rough-handling test end points					

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Quality conformance inspection, part 1					
Appendix D, 20(b)	Dimensions	See figures 1, 2, and 3				
1301	Heater current	Ef = 4.75V; Note 1	If	48	58	A
1256	Cathode current	Ef = 4.75V; peak beam voltage = 125 kv	ib	84	100	a
1231	Emission	Note 3	Δib/ib	---	5	%
----	Power output (1)	peak beam voltage = 128 kv; Ef = 4.75V; pd = 2 kw; F = 5650 Mc; Magnet coil current = 11.5A; Notes 6 (b), (c), (d), 8(a), (b), (e), 10(d), and 12(b), (d)	po 1	2.7	---	Mw
----	Power output (2)	peak beam voltage = 128 kv; Ef = 4.75V; pd = 10 kw; F = 5650Mc; Magnet coil current = 12A; Notes 6 (b), (c), (d), 8(a), (c), (e), 10(d), and 12(b), (d)	po 2	3	---	Mw
	Quality conformance inspection, part 2					
4218	Tunable frequency	po 3; Notes 4 and 8	F	5400	5900	Mc
1331	Direct interelectrode capacitance	Electron gun to tube body	---	---	40	pf
----	Power output (3)	peak beam voltage = 128 kv; Ef = 4.75V; F = 5400 to 5900 Mc (frequency shall not vary more than 100 Mc); pd = 10 kw; Magnet coil current = 12A; Notes 6(b), (c), (d), 8(a), (c), (d), (e), 10(d), 12(b), (d)	po3	3	---	Mw
----	X-Radiation	Ef = 4.75V; peak beam voltage = 130 kv; pd = 10 kw; po = 3.15 ± 0.15 Mw; Magnet coil current = 12A; F = 5400 Mc; Notes 6(b), (c), (d), 8(a), (c), (e), 10(d), 12(b), (d), and 17	XR1 XR2 XR3	--- --- ---	8 50 30	mr/hr mr/hr mr/hr
	Quality conformance inspection, part 3					
4.7	Life-test provisions	peak beam voltage = 128 kv; Ef = 5V; pd = 10 kw; F = 5650 Mc; Magnet coil current = 12A; Notes 6(b, c, d) except flat top, 8(a), (c), (e), 10(d), and 12(a), (b), (d)	t	500	---	hrs
4.7.3	Life test end point	po 2	po	2.7	---	Mw

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NOTES:

1. When first applying heater voltage, heater surge current shall not be allowed to exceed 70 amps at any time. Full heater voltage of 4.75 volts shall be applied for 5 minutes before any other voltages are applied. The heater current shall stabilize to the limit specified herein within 5 minutes.
2. New tubes, or tubes stored over 3 months shall require special start-up procedure as follows:
   (a) Heater shall be warmed-up in accordance with note 1.
   (b) After heater warm-up, apply approximately a peak beam voltage of 50 kv for 30 minutes. then apply approximately a peak beam voltage of 100 kv for 30 minutes, after which the beam voltage shall be raised to that required for 3 Mw output under the conditions specified for power output tests.
3. Under the conditions specified herein for Cathode current, Ef shall be decreased from 4.75 to 4.25 volts. The percentage change in cathode current shall be less than the limit specified herein and is defined as emission.

$$\frac{ib\ (Ef = 4.75\ V) - ib\ (Ef = 4.25\ V)}{ib\ (Ef = 4.75\ V)} \times 100 = \Delta \frac{ib}{ib} \times 100$$

4. (a) A plasticized tuning curve shall be supplied with each tube. The frequency of each resonator shall be within ± 25 Mc/sec. at the settings specified in the tuning charts. Full tuning of input and output resonators shall be accomplished in 22 ± 3 turns and full tuning of the middle resonator shall be accomplished in 24 ± 4 turns.
   (b) The torque required to adjust the tuners shall not exceed 5 in-lbs at any point in the tunable frequency range.
   (c) Tuners shall remain undamaged with a torque of 10 in-lbs applied to the tuner shaft at the extreme tuner positions.
5. (a) Liquid coolant shall be supplied to the drift tube and collector. The coolant connectors shall be corrosion resistant steel coupling units, AN 818-6 on drift tube and AN 818-10 on collector.
   (b) Drift tube heating varies with focus coil current, beam voltage, r.f. drive power applied and tuning of middle and output resonators. Flows at inlet temperatures listed for collector and drift tube will allow safe operation for the conditions of this specification. Increasing flow or decreasing inlet temperature (inlet temperature shall remain above minimum listed values) will also provide safe operating conditions. Flows for inlet temperatures between values listed may be interpolated by assuming a linear relationship between any two adjacent values.
   (c) Absolute maximum inlet temperatures specified herein shall not be exceeded under any conditions of operation of the tube. Recommended operating maximum values provide a small margin of safety for short term fluctuations in flow rate or inadvertent short term operation of the tube outside this specification. Flows below absolute minimums listed herein may fail to provide heat transfer and shall not be permitted under any conditions of operation. Temperatures below 10°C are listed for operational information and shall not be tested by the manufacturer. Pressure drops listed herein are for flow at temperature listed.

Coolant - Water					
Condition	Inlet Temp. °C	Collector		Drift Tube	
		Flow gpm Min.	Pressure Drop Psi Maximum	Flow gpm Min.	Pressure Drop Psi Maximum
Abs. Max. (T)	77	9.2	46-1/2	3.0	46-1/2
Recommended Operating Max. (T)	70	6.0	24	1.45	21-1/2
	65	5.0	17	1.15	11-1/2
Abs. Min. (flow)	62	---	--	1.0	7-1/2
	56-1/2	4.0	12	1.0	7-1/2
	0 to 40	3.0	8	1.0	7-1/2

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NOTES – Continued:
5. Continued:

Coolant - 50% Ethylene Glycol (Vol.)					
Condition	Inlet Temp. °C	Collector		Drift Tube	
		Flow gpm Min.	Pressure Drop Psi Maximum	Flow gpm Min.	Pressure Drop Psi Maximum
Abs. Max. (T)	67	6.6	36	3.0	54-1/2
Recommended Operating Max. (T)	65	6.0	32-1/2	2.25	36-1/2
	57	5.0	26	1.25	15
Abs. Min. (flow)	-32 to +35	3.8	18	1.25	15

(d) A minimum absolute pressure of 35 psia shall be maintained at the outlet of all water cooling passages of the tube and magnet.
(e) Storage of tubes at temperatures below 1°C (water) or -32°C (50 percent ethylene glycol) with liquid in the coolant passages is prohibited as construction is such that damage to the tube may result in lower temperatures.

6. (a) Beam pulse width shall be measured between the 50 percent points of the beam current pulse when viewed by a linear circuit which presents a voltage representation of the pulse.
(b) With 12.5 amperes d.c. field current applied, the axial gauss level measured on the coil axis 8 inches from the top surface of the structure shall be 1245 ± 95 gausses. Tests shall be performed with the following beam pulse characteristics: (Times listed are nominal).

Characteristics	Description	Defines
Pulse top[1/]	Flat ± 2 percent for 3.0 μs	peak beam voltage
Rise time	0.6 μs (10-90 percent)	peak beam current
Decay time	1.8 μs (90-10 percent)	peak beam current

[1/] Pulse top is considered the 100 percent point and is defined as the arithmetic mean of amplitude variations across the top of the pulse. All listed values of peak beam voltage shall be determined on this basis.

(c) The r.f. pulse width shall be measured between the 3 db points of the r.f. output pulse when viewed through a crystal detector or equivalent circuit. There shall be no random variations in the r.f. pulse width not attributable to a source external to the tube (that is, beam pulse, r.f. drive pulse, and so forth).
(d) The maximum operational prf shall be determined by the maximum duty cycle and the pulse width applied.

7. An external circuit shall be provided to limit the momentary cathode current to a maximum value of 120 amperes. A 1/4 inch minimum thickness lead cylinder shall be placed between the inner core of the structure and the field winding. This lead cylinder shall not be any portion of the 9.200 inner diameter.

8. (a) The power output shall be measured on the basis of the output r.f. pulse width.
(b) The tube shall have all resonators tuned identically to the drive frequency at the peak beam voltage in accordance with Note 12(d). With 2 kw drive power, the middle resonator shall then be tuned to the high frequency side of the drive frequency until output is maximized.
(c) The tube shall have all resonators tuned identically to the drive frequency at the peak beam voltage in accordance with note 12(d). The middle resonator shall then be tuned to the high frequency side of the drive frequency by 100 Mc per second (approximately four turns). The drive shall then be increased to 10 kw after which the middle resonator shall be adjusted until the desired output power is obtained. In some cases, optimum tuning of the middle resonator will provide output powers in excess of the 3.3 Mw maximum rating and a detuned setting of the middle resonator only will be required for operation.

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NOTES — Continued:

8. Continued:
   (d) At operational peak beam voltage, the VSWR looking into the input connector of the tube shall be 1.8:1 or less, over the frequency range under po 3 conditions.
   (e) Manufacturer's final test shall be performed at the peak beam voltage specified herein. For retest at other than the manufacturer's facility, the minimum values of po 1 and po 2 specified herein shall be obtained at a beam voltage value not to exceed the 135 kv specified herein with a known accuracy of ± 5 percent.

9. (a) The tube shall have all resonators tuned identically to the drive frequency at the peak beam voltage in accordance with note 12(d).
   (b) For all values of pd between 1 and 100 watts, the gain shall be as specified herein.

10. (a) The power output shall be as specified in power output test conditions for any phase of a 1.3:1 VSWR load.
    (b) The power output shall be not less than 2.6 Mw for any phase of a 1.5:1 VSWR load under po 2.
    (c) The output window of the tube shall be pressurized to 45 psia (minimum) whenever r.f. drive power and beam voltage are simultaneously applied.
    (d) The tube shall not be damaged by a load under the following conditions:

VSWR (any phase)	Output waveguide pressure (p.s.i.a.)
1.5:1 (or less)	46.5 ± 1.5
1.7:1	56 ± 2
---	65 (pressure test)

11. (a) The maximum r.f. drive pulse length shall be 6 μs (3 db point when viewed through a crystal detector operating in the linear region or an equivalent circuit).
    (b) The tube shall not be damaged when operated with maximum r.f. drive power when the peak beam voltage = 0 volt.

12. (a) The tube shall be operated as shown on figure 4, Magnet focusing structure.
    (b) Optimum magnet current will vary somewhat from tube to tube and, therefore, shall be optimized for each tube. The optimum value of current shall be within ± 5 percent of the values specified herein under power output test conditions.
    (c) The minimum safe magnet current for all conditions of this specification is 11.2 amperes, except as noted for po 1 and low level gain conditions. However, as the safe magnet current depends on the beam voltage, drive power, and tuning of the tube, this is not necessarily a safe value for all possible operating conditions outside of this specification. Under no conditions shall the tube be operated at rated voltage with less than 10 amperes field current.
    (d) Tuning of all resonators identically to the drive frequency shall be accomplished with 11.0 amperes magnet current, with less than 750 w drive power applied.

13. With the heater applied for at least 5 minutes, the klystron output shall be at least 3 Mw within 10 seconds after beam voltage is applied.

14. The electron gun insulator of the tube shall be operated in oil. Univolt 35, or equivalent grade oil shall be used.

15. Two thermal safety switches, monitor temperature of the collector and drift tubes. Except as specified herein a short shall exist between the interlock terminals. Under no conditions can the tube be operated in a closed coolant circuit with an external short applied between the interlock terminals. Switches are calibrated to open at 240°F. (115.5°C) ± 5°F., close at 220°F (104.5°C) ± 7°F. Ratings of switches are 3 amps at 110 volts A.C. Opening and closing temperatures shall be certified by the thermal switch manufacturer. Verification tests shall be performed on the thermal switch only.

16. Tubes meeting the requirements specified herein for Emission and Tunable frequency, after this test, shall be considered as meeting the requirements of this test.

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NOTES – Continued:

17. (a) The X-radiation shall not exceed the limits specified when the tube is operated under the conditions specified herein.
    (b) XR_1 shall be measured at a distance of 1 foot in all directions from the outer surfaces of the collector lead shield above the magnet structure mounting flange, except for the direction and area controlled by XR_2.
    (c) XR_2 shall be measured at a distance of 1 foot from the output waveguide mounting flange on a line perpendicular to the flange face passing through the center of the flange. The cross sectional area of the radiation pattern covered by XR_2 shall be less than twice the maximum internal dimension of the waveguide squared.
    (d) XR_3 shall be measured at distance of 1-1/2 feet in all directions normal to the outer surface of the magnet structure below the magnet structure mounting flange.

18. When the tube is operated under po 2 conditions, variations in inlet coolant temperature within the range of 10° to 70°C. shall not result in power output variations in excess of the limit specified herein.

19. This is a destructive test. The tube manufacturer may use tubes in the performance of this test that do not meet all production specifications contained herein, except tubes that do not meet the frequency range specified herein shall not be used.

20. The tube shall withstand the shock test specified in MIL-STD-202, Method 205, test B. The shock test shall consist of six shocks, two in each of three mutually perpendicular planes, or equivalent. The tube shall be mounted in a shock-mounted fixture such that the tube and mounting fixture have a natural frequency of 31 ± 3 cps with a maximum deflection of 3/8 ± 1/16 inch. Tubes meeting the requirements specified herein for Emission and Tunable frequency, after this test, shall be considered as meeting the requirements of this test.

21. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

22. Average beam power, as determined by the product of the average beam current and the peak beam voltage or the product of the peak beam current, peak beam voltage and the beam DU, shall not exceed the maximum average beam power specified herein.

23. CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC. PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N705(NAVY))

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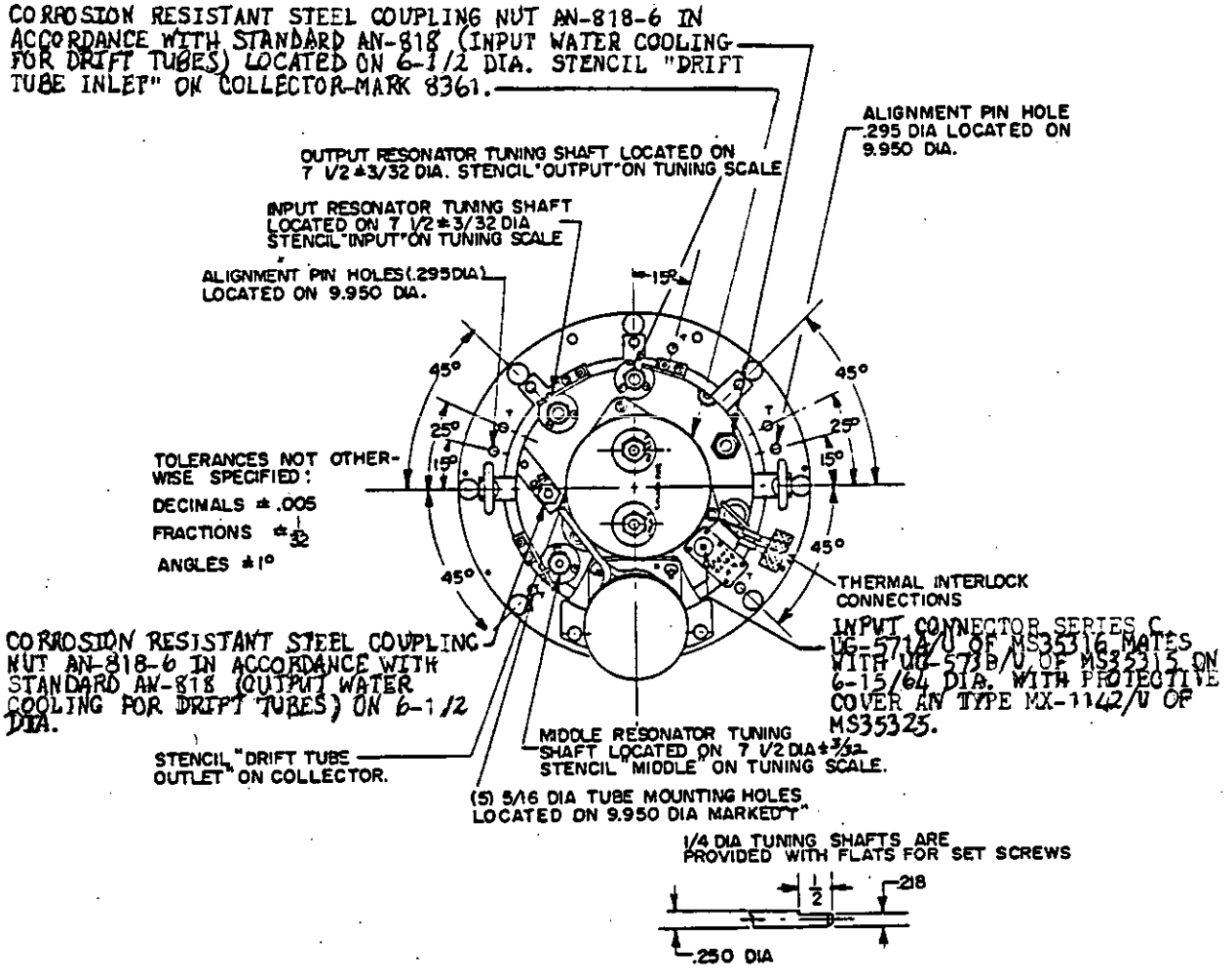


Figure 1 - Collector end dimensions of tube type 8361.

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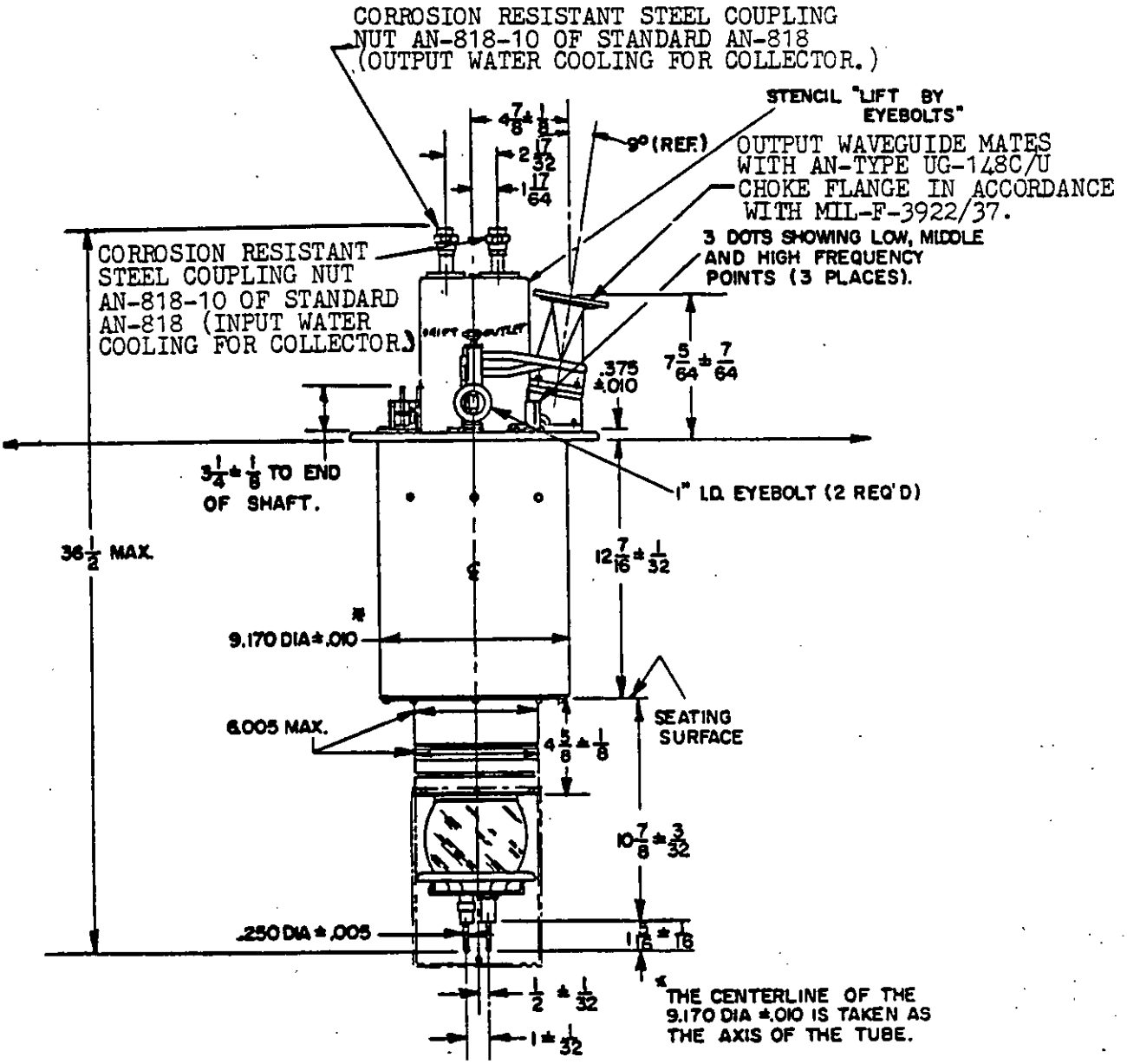


Figure 2 - Dimensions of tube type 8361.

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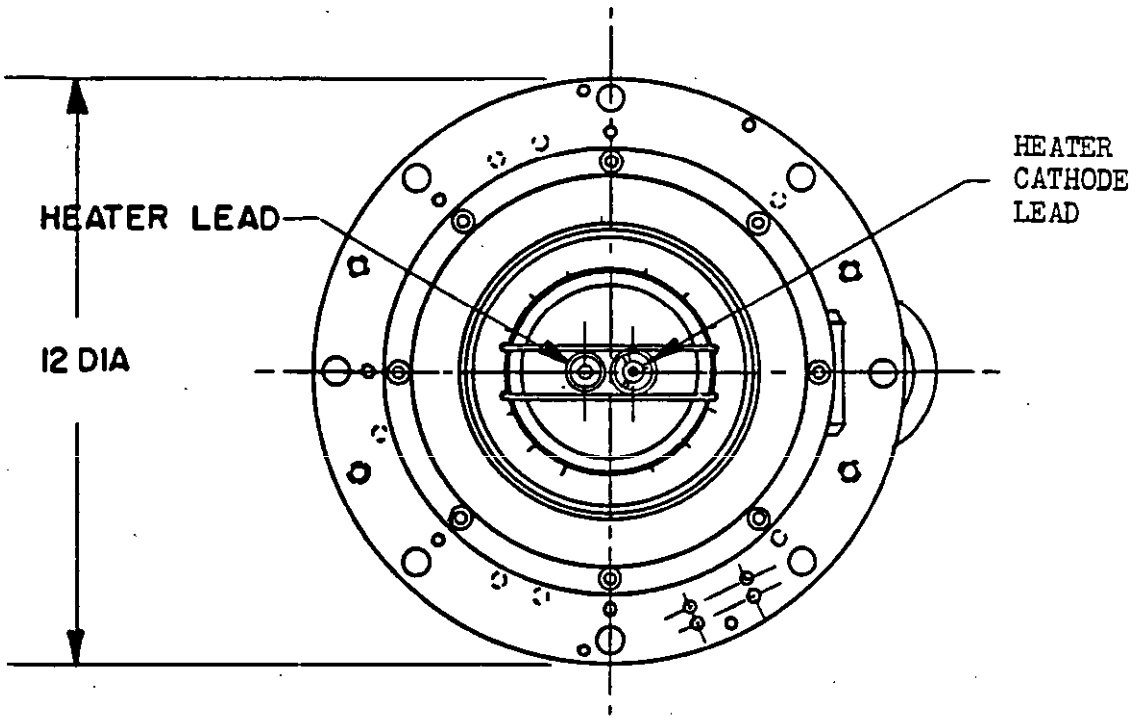
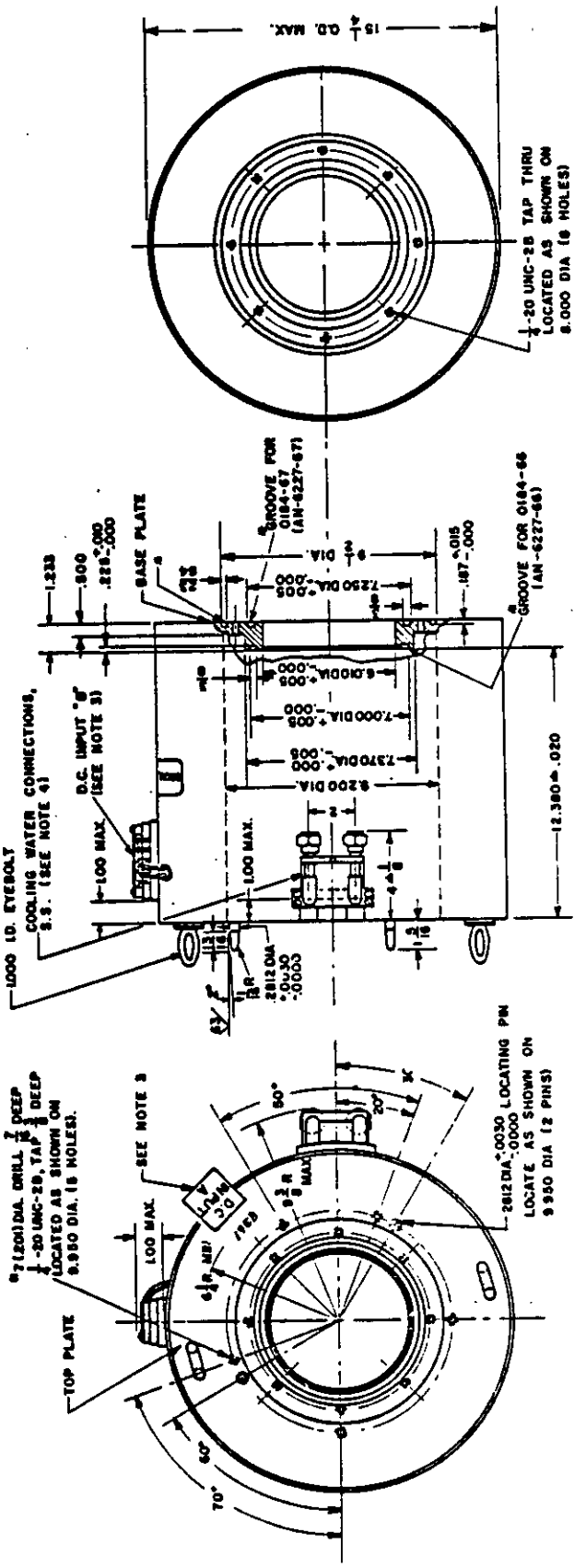


Figure 3 - Cathode end dimensions of tube type 8361.

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NOTES:

1. Structure and operating characteristics shall be as specified herein.
2. #63/finish required on internal groove surfaces.
3. Magnets shall be supplied with d.c. input located in position "A" unless otherwise specified in the contract or purchase order.
4. Cooling water connections shall be AN-818-6 unless otherwise specified in the contract or purchase order.
5. Dimensions are in inches unless otherwise specified herein. Tolerances are as follows unless otherwise specified herein.

Decimals	± 0.005
Fractions	± 1/16
Angles	± 1 degree

Figure 4 - Magnet focusing structure.

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